

A Zero-Centered Lag-1 Diagnostic for Higher-Order Differences of the Primes

Independent Technical Note

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Abstract

We introduce a scalar diagnostic δ on the sequence of prime numbers. Let $g_n = p_{n+1} - p_n$ be the ordinary prime gaps and form the second differences $\Delta^2 g_n = \Delta^3 p_n$. Compute the centered lag-1 sample autocorrelation ρ_1 of this series and define

$$\delta := \rho_1 + 2/3.$$

The constant $-2/3$ is the exact theoretical lag-1 autocorrelation acquired by white noise after two successive differencings. Consequently δ measures excess (or deficit) serial correlation relative to a pure white-noise null on the gaps. We give a rigorous derivation of the floor, report numerical values on the first 10^5 primes, and supply minimal reproducible code. This note establishes priority for the diagnostic and its centering convention.

1. Definition

Let $(p_n)_{n \geq 1}$ be the sequence of primes and $g_n = p_{n+1} - p_n$. The second forward differences of the gaps are

$$D_n := \Delta^2 g_n = g_{n+2} - 2g_{n+1} + g_n = \Delta^3 p_n.$$

Given a finite initial segment $\{D_1, \dots, D_N\}$ with $N \geq 2$, form the centered lag-1 autocorrelation

$$\rho_1 = c_1 / c_0,$$

where

$$c_0 = (1/N) \sum (D_k - D_{\blacksquare})^2, \quad c_1 = (1/(N-1)) \sum (D_k - D_{\blacksquare})(D_{k+1} - D_{\blacksquare})$$

and $D_{\blacksquare}$ is the sample mean. The diagnostic is then

$$\delta := \rho_1 + 2/3.$$

The same construction applies verbatim to any real sequence in place of the primes.

2. Derivation of the White-Noise Floor

Let ε_t be white noise with $E[\varepsilon_t] = 0$, $\text{Var}(\varepsilon_t) = \sigma^2 < \infty$ and $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$.

First difference: $y_t = \varepsilon_t - \varepsilon_{t-1}$. $\text{Var}(y_t) = 2\sigma^2$, $\text{Cov}(y_t, y_{t-1}) = -\sigma^2$, hence $\rho_1(y) = -1/2$.

Second difference: $z_t = y_t - y_{t-1} = \varepsilon_t - 2\varepsilon_{t-1} + \varepsilon_{t-2}$. Expanding the second-moment expressions yields

$$\text{Var}(z_t) = 6\sigma^2, \quad \text{Cov}(z_t, z_{t-1}) = -4\sigma^2.$$

Therefore $\rho_1(z) = -4/6 = -2/3$.

Thus, if the gaps g_n themselves behaved as white noise, the series $D_n = \Delta^2 g_n$ would satisfy $\rho_1 = -2/3$ and $\delta = 0$ in the large-sample limit. The additive shift by $+2/3$ centers the diagnostic on this null. (The same calculation shows that the k -th difference of white noise has lag-1 autocorrelation $-k/(k+1)$.)

3. Numerical Results

On the first $N = 10^5$ primes (no preprocessing) one obtains

$$\rho_1 \approx -0.67439, \quad \delta \approx -0.00773.$$

The deviation from the theoretical floor is therefore of order 10^{-2} . Optional linear detrending followed by winsorization at the 1%/99% quantiles shifts ρ_1 upward (toward -0.5). Phase-randomized and shuffle surrogates remain tightly concentrated about $-2/3$, confirming that the observed δ is not an artifact of the particular finite window.

4. Relation to Nearby Observations

Independent analyses have reported lag-1 sample autocorrelations clustering near -0.52 when second differences of prime gaps are examined. Those measurements concern the same series D_n that appears in the definition of δ . A value near -0.52 lies between the pure first-difference floor ($-1/2$) and the pure second-difference floor ($-2/3$). The present construction simply recenters the same autocorrelation by the exact second-difference constant, converting a raw coefficient into a signed excess relative to the white-noise null on the gaps. No stronger claim of identity is made; the two quantities differ in centering convention and in sample-size and preprocessing choices.

5. Minimal Reproducible Implementation

The following Python code implements the canonical path (primes $\rightarrow$ gaps $\rightarrow$ second differences $\rightarrow \rho_1 \rightarrow \delta$):

```
import numpy as np

def compute_delta(primes):
    g = np.diff(primes)          # first differences (gaps)
    D3 = np.diff(g, n=2)         # second differences of gaps
    x = D3 - D3.mean()
    c0 = np.dot(x, x) / len(x)
    c1 = np.dot(x[:-1], x[1:]) / (len(x) - 1)
    rho1 = c1 / c0
    return {"rho1": float(rho1),
            "delta": float(rho1 + 2.0/3.0),
            "n": int(D3.size)}

# Example:
# p = np.array([...]) # first N primes
# print(compute_delta(p))
```

6. Remarks and Priority Statement

The diagnostic is deliberately elementary. Its principal virtues are (i) an exact, distribution-free null value derived from second-moment calculations alone, and (ii) immediate computability on any ordered sequence. Whether the small negative δ observed for the primes is ultimately significant, or merely a finite-sample or arithmetic-progression artifact, remains open; the construction itself supplies a clean baseline against which sharper tests can be compared.

This note is deposited to establish a public, timestamped record of the definition of δ , the centering by $+2/3$, the associated white-noise derivation, and the minimal computational procedure. All rights to further development, software implementations, and extended analyses are retained by the author.